

Our Atmosphere is How Big?

Comparing the Layers

Earth's **exosphere** (the uppermost layer of the atmosphere) is the largest layer and extends from 700 km above sea level to about *10,000 km* above sea level. The upper boundary of the exosphere is a **transitional zone** between Earth's atmosphere and space. Molecules in the exosphere are **gravitationally bound** to Earth but are so far apart that they do not behave as a gas. Because the exosphere is so large and empty, we typically look at Earth's atmosphere from the lower boundary of the exosphere down.

The **thermosphere** is the second highest layer of the atmosphere, and the second largest. If you were to scale the atmosphere in a typical room where the upper boundary of the thermosphere is the ceiling, the lower boundary would only be 10 inches off the floor!

Directions

This activity will have you create a scale diagram of Earth's atmosphere from the surface of the Earth to the lower boundary of the thermosphere.

You will be using the following formula to calculate scaled altitude:

$$\text{Scaled Altitude in centimeters} = (a \text{ km}) \times (23 \text{ cm} / 85 \text{ km})$$

Use the scaled altitude calculations to draw a diagram of the lower atmosphere in your notebook. You may use a calculator to perform the calculations. Round scaled altitude to the nearest mm (tenth of a centimeter). Use the vertical red line as your Y axis. Have the second blue line from the bottom be 0km (sea level). Be sure to label each feature and label the vertical axis with the actual altitudes of each feature.

Feature	Altitude (<i>a</i>)	Scaled Altitude (cm)
<i>Mesosphere</i>	<i>50-85 km</i>	
<i>Stratosphere</i>	<i>10-50 km</i>	
<i>Troposphere</i>	<i>0-10 km</i>	
Felix Baumgartner	39 km	
Mount Everest	8.8 km	
Tallest Building	0.8 km	
<i>Ozone Layer</i>	<i>15-35 km</i>	
Commercial Jets	8-11 km	

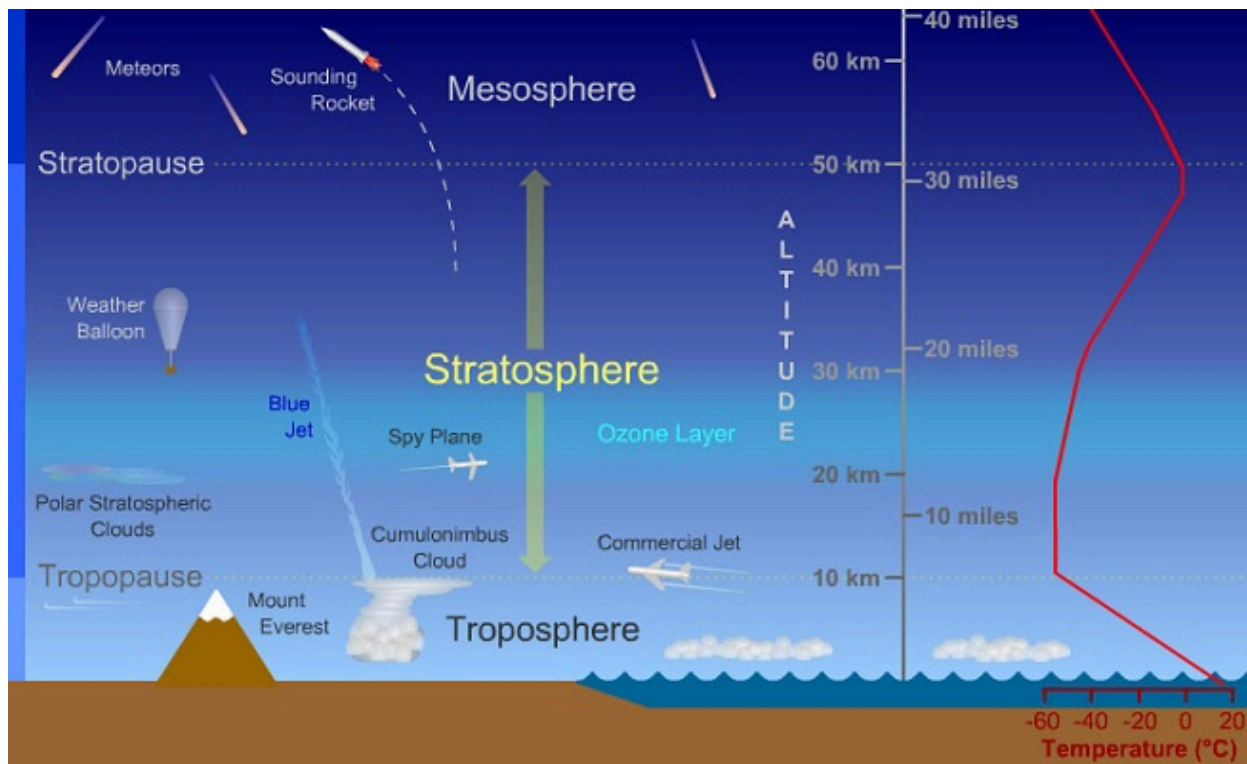
Meteors Burn	50-95 km	
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**Italicized features are layers/zones: draw boundary lines at the min & max altitude.*

Non-italicized features that exist within a range should be drawn in that range.

Discussion Questions:

Use the image to answer the following questions in your notebook.



Questions:

1. What is the lowest layer of the atmosphere?
2. What layer of the atmosphere contains the ozone layer?
3. What layer of the atmosphere contains most clouds?
4. What layer of the atmosphere do meteors burn up in?
5. In the troposphere, does temperature decrease or increase as altitude increases?
6. In the stratosphere, does temperature decrease or increase as altitude increases?
7. What layer of the atmosphere did Baumgartner jump from? Why didn't he burn up like a meteor does?